

Hooman Ramezani

hooman125@gmail.com | www.hoomanramezani.com | github.com/HoomanRamezani

EDUCATION

University of Toronto <i>MASc in Machine Learning & Operations Research</i>	Toronto, ON 2022 – 2024
Thesis: Transformers for clinical workloads; LLM and ViT models with multimodal data for lung cancer treatment	
University of Waterloo <i>BASc in Systems Design Engineering</i>	Waterloo, ON 2017 – 2022
GPA: 3.7 / 4.0 — Coursework: Intro. Deep Learning, Intro. Machine Learning, Pattern Recognition, Neuroscience	

EXPERIENCE

AI Specialist Solutions Architect <i>Nebius Group (Neocloud)</i>	Jan 2025 – Present Remote
- Building optimized large scale LLM training/inference solutions on H100/B200/GB200 with NVLink + IB - Performance research using Nsight Systems/DCGM/PyTorch Profiler to identify training bottlenecks - Implement FSDP/TP/PP, efficient dataloadings pipelines to raise MFU for customers - Stand up large GPU clusters for clients (1000+ GPUs) with slurm / k8s, WEKA storage, o11y	
Machine Learning Researcher <i>University Health Network</i>	May 2023 – Dec 2024 Toronto, ON
- First-author ‘LN-Transformer’ paper for lung nodule segmentation, achieving DiceC of 94.2% (SOTA) - Built multi-modal LLM with unstructured text, CT slices for treatment recommendations with 84% concordance	
Machine Learning Engineer <i>Advanced Micro Devices (AMD)</i>	Sept. 2022 – April 2023 Toronto, ON
- Optimized LLM inference on AMD CPUs via QAT, post-quantization, pruning, and knowledge distillation, optimized GPU kernel execution to reduce inference latency by 80% and maximize throughput - Benchmarked against Triton Inference Server with TensorRT backend on k8s, leveraging dynamic batching	
Machine Learning Engineer <i>DarwinAI (Acquired by Apple)</i>	Jan. 2022 – April 2022 Remote
- Delivered medical vision model to Pfizer for advanced liver fibrosis diagnosis, achieving 91% sensitivity - Built end-to-end ML solutions to clients with MLOps CI/CD pipelines, enabling continuous retraining, monitoring	

RESEARCH // PROJECTS

LN-Transformer: Lung Nodule Transformer for Sparse CT Segmentation	2025
- Published a novel two-stage transformer for lung nodules segmentation featured in CVPRW - Strongest model on benchmark dataset Dice 91.4%, F1 94.2%, includes Meta SAM, DETR architectures	
Livy Education Chatbot Agent <i>GPT-4, RAG, LangChain, Chroma</i>	2024
- Developed a RAG assignment research chatbot for Canvas students, GPT-4, Chroma, RAG reranking - Built context-aware AI agent to adaptively monitor student engagement and provide real-time support	
Enhanced Robotic Grasping with PointNet <i>LiDAR, Robotics, 3D Computer Vision</i>	2022
Trained PointCloud grasp proposition neural-net achieving 87% grasp success rate, identify optimal grasp	

TECHNICAL SKILLS

Programming: Python, C/C++, Java, SQL, JavaScript
ML & DL: PyTorch, TensorFlow, Hugging Face, LoRA / PEFT, RAG, AI Agents, TensorRT, ONNX, Triton, CUDA
Distributed Systems: PyTorch DDP, FSDP, Spark, Kafka, Airflow, Docker, Kubernetes, Slurm
Cloud & MLOps: AWS (S3, EC2, Lambda, SageMaker), GCP, MLflow, Prometheus, Redis, Flask, Git